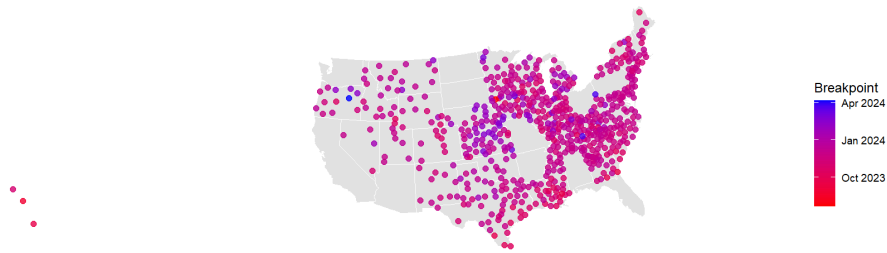
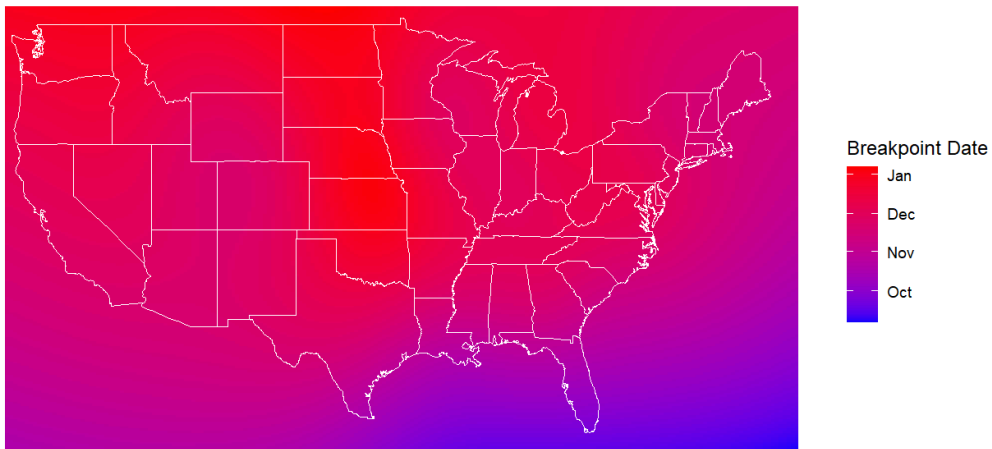


## INFLUENZA



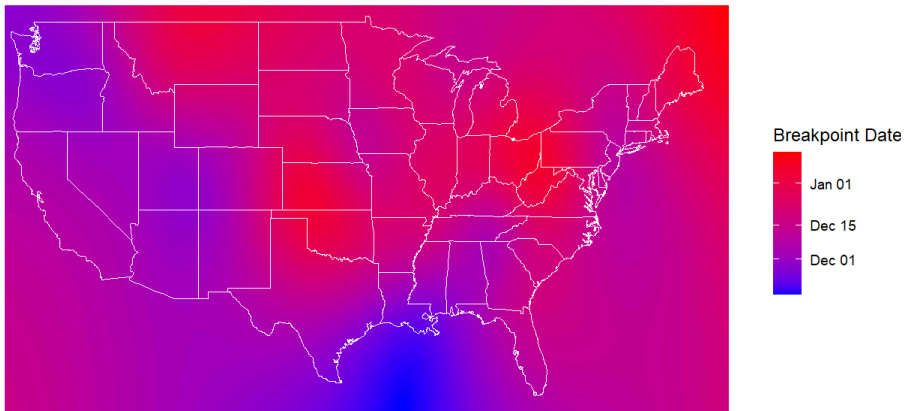
**map\_2023\_2024\_influenza:** no data from California, Florida, North/South Dakota, Ohio, Pennsylvania, New Mexico, Washington, Arkansas, Missouri, Alabama

Smoothed Influenza Breakpoint Dates across U.S. HSAs



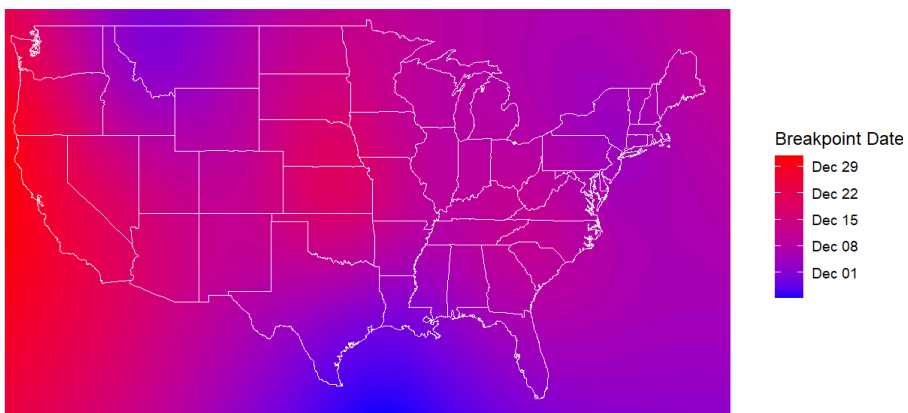
**interpol\_2023\_2024\_influenza:** outbreaks started at the southeast

Smoothed Influenza Breakpoint Dates across U.S. HSAs

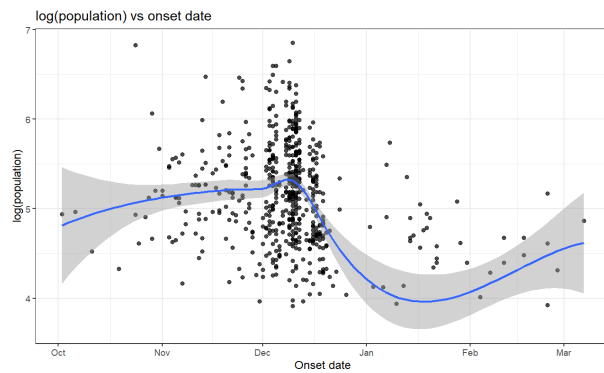
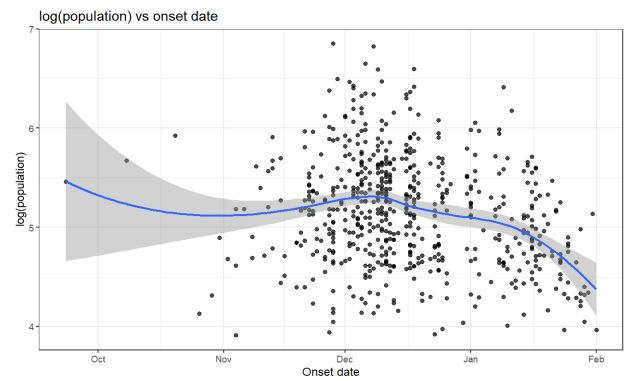
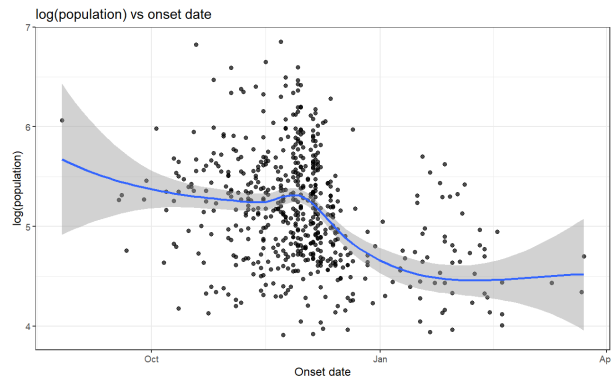


**interpol\_2024\_2025\_influenza:** outbreaks started at the south and west

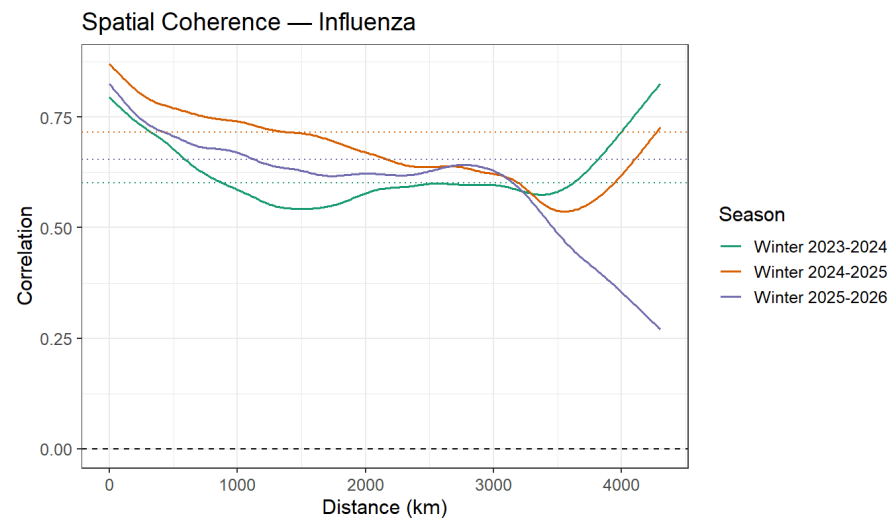
Smoothed Influenza Breakpoint Dates across U.S. HSAs



**interpol\_2025\_2026\_influenza:** outbreaks started at the east and moved to the west



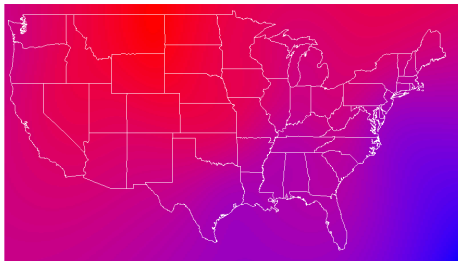
**logpop\_onset\_2023\_2024\_influenza, logpop\_onset\_2024\_2025\_influenza, logpop\_onset\_2025\_2026\_influenza:** regions with larger population size tend to experience outbreaks earlier, no such pattern for regions with smaller population size



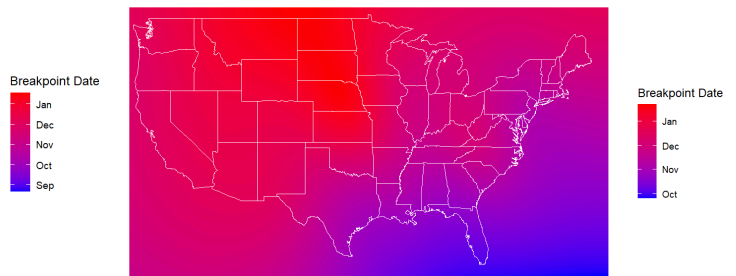
**spatial\_coherence\_superimposed\_influenza\_nohawaii:** east and west coasts are synchronized during winters 2023-2024 and 2024-2025, more diffusion in winter 2025-2026

## RSV

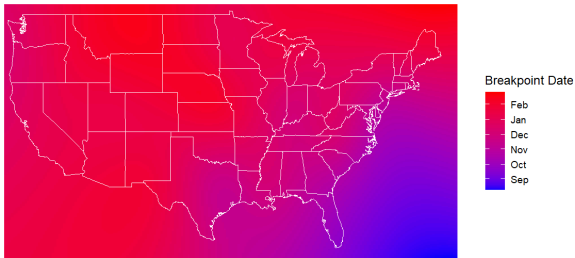
Smoothed RSV Breakpoint Dates across U.S. HSAs



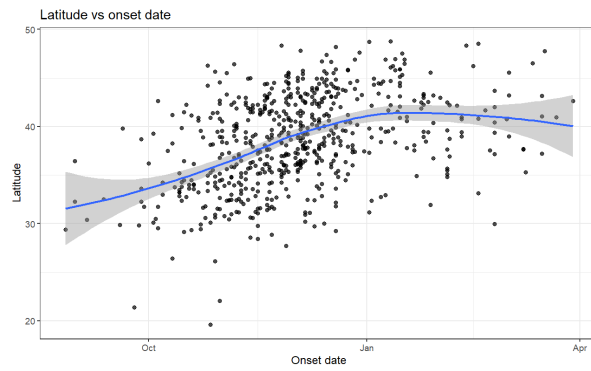
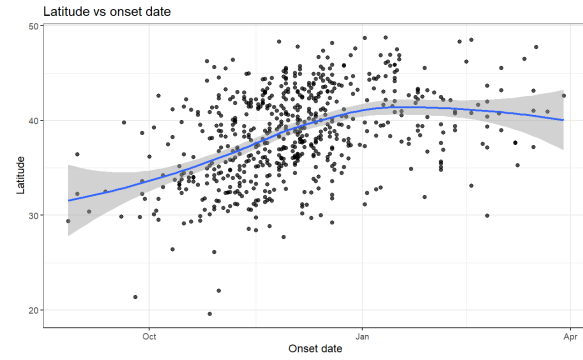
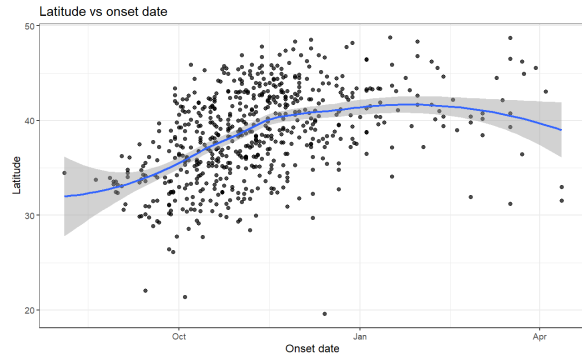
Smoothed RSV Breakpoint Dates across U.S. HSAs



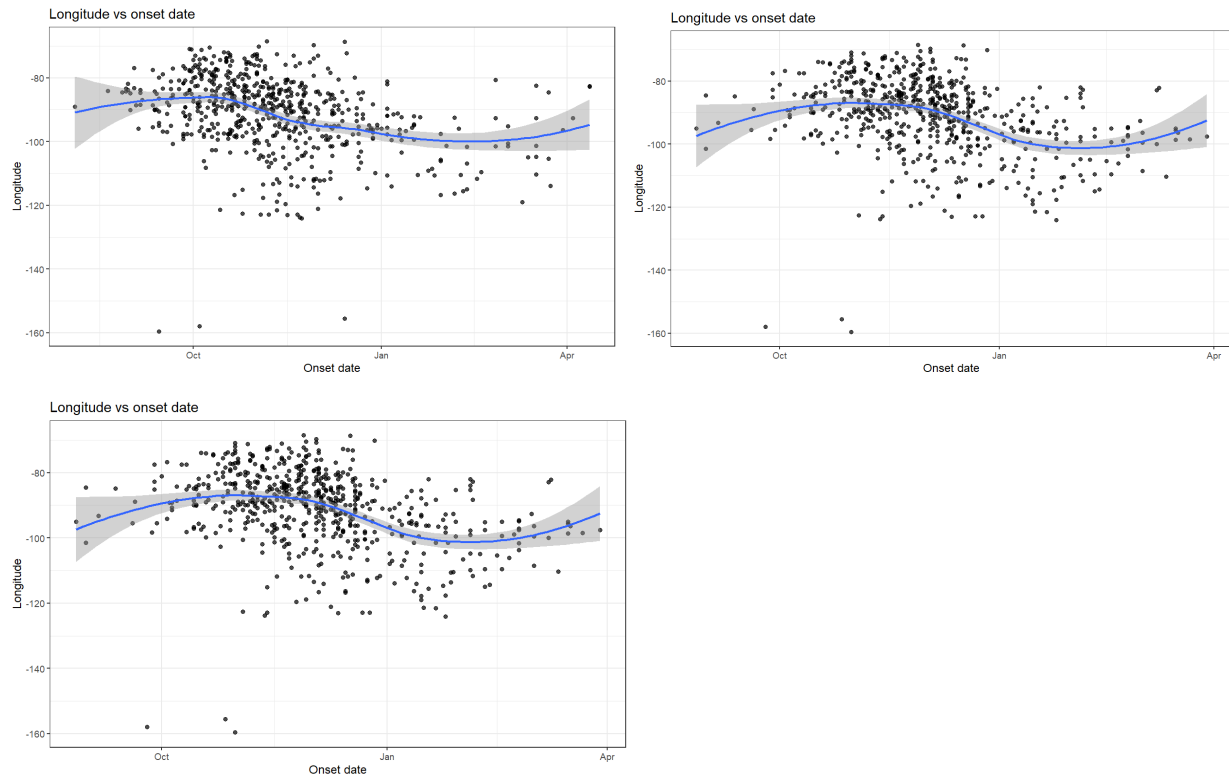
Smoothed RSV Breakpoint Dates across U.S. HSAs



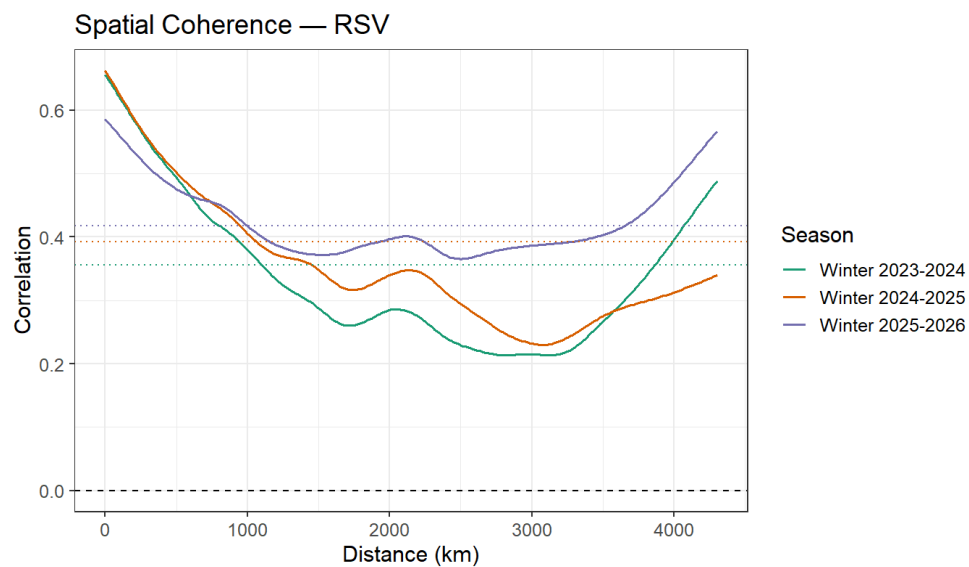
**interpol\_2023\_2024\_rsv, interpol\_2024\_2025\_rsv, interpol\_2025\_2026\_rsv:** outbreaks start in the southwest in all years



**lat\_onset\_2023\_2024\_rsv, lat\_onset\_2024\_2025\_rsv, lat\_onset\_2025\_2026\_rsv:** southern regions tend to experience outbreaks earlier

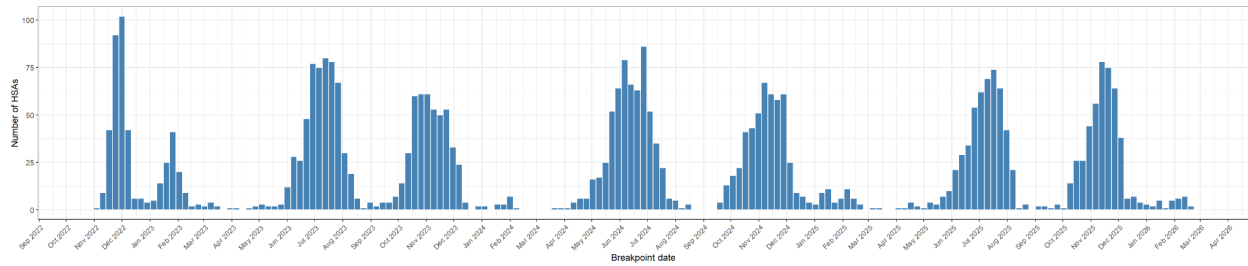


**lon\_onset\_2023\_2024\_rsv, lon\_onset\_2024\_2025\_rsv, lon\_onset\_2025\_2026\_rsv:**  
westernmost regions tend to experience earlier outbreaks



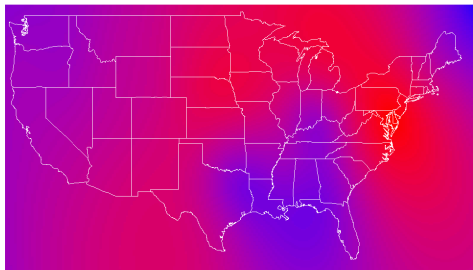
**spatial\_coherence\_superimposed\_rsv\_nohawaii:** eastern and western regions are synchronized for all winters

## COVID

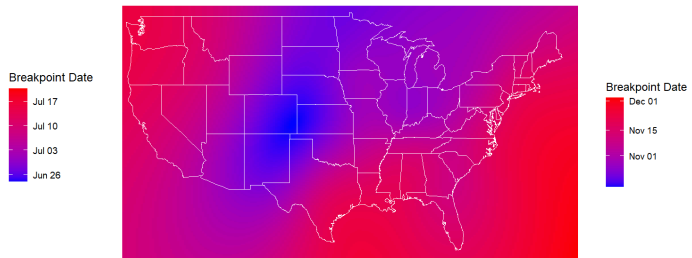


**histogram\_covid:** to separate the outbreak waves

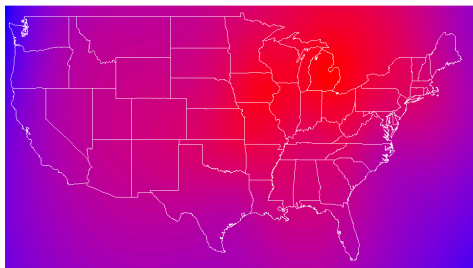
Smoothed COVID Breakpoint Dates across U.S. HSAs



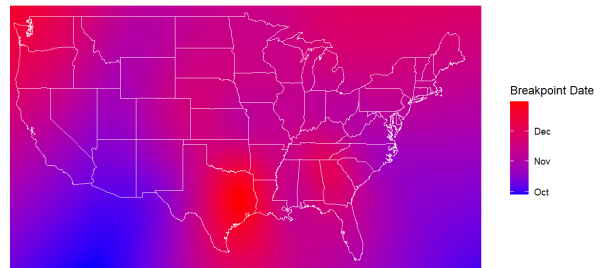
Smoothed COVID Breakpoint Dates across U.S. HSAs



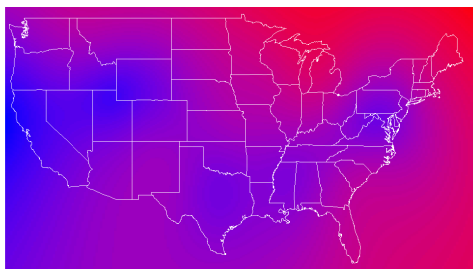
Smoothed COVID Breakpoint Dates across U.S. HSAs



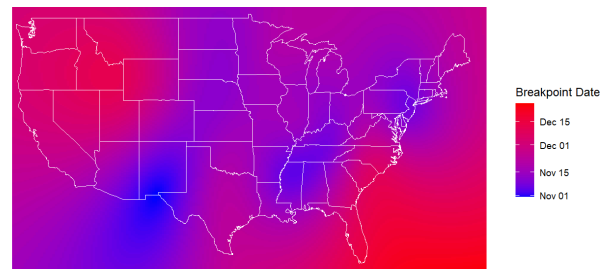
Smoothed COVID Breakpoint Dates across U.S. HSAs



Smoothed COVID Breakpoint Dates across U.S. HSAs

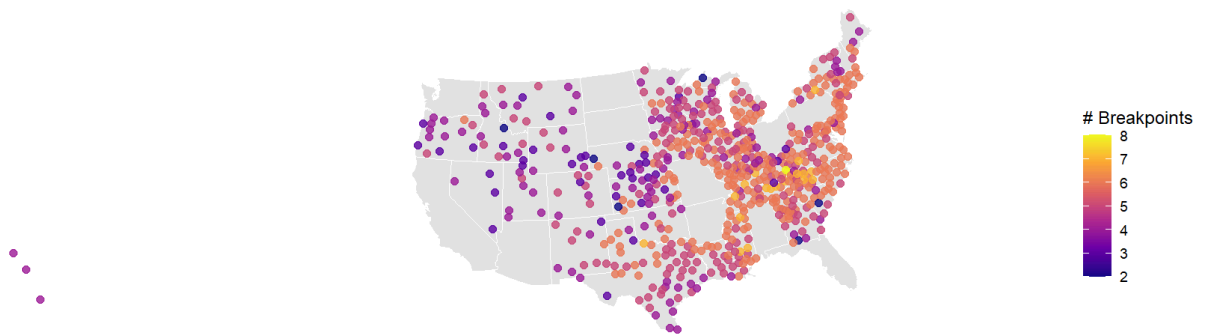
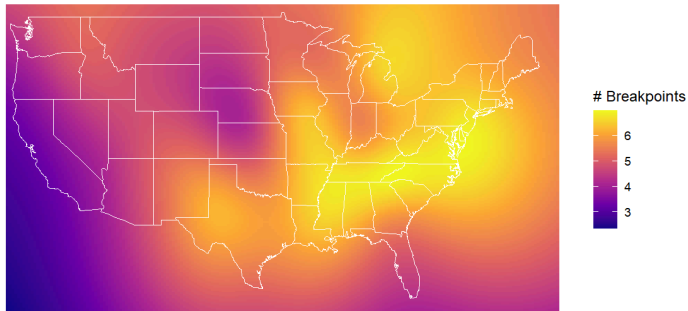


Smoothed COVID Breakpoint Dates across U.S. HSAs

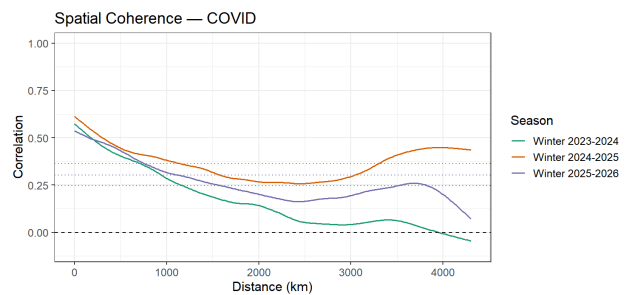
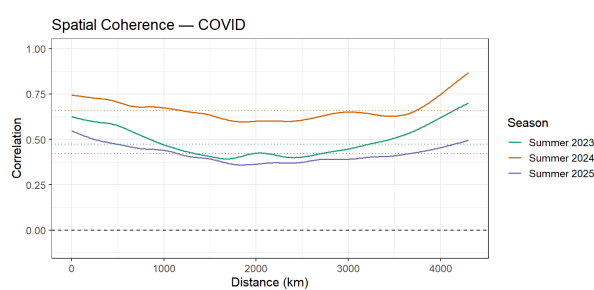


**COVID interpolations in chronological order (interpol\_2023-03-20\_2023-09-01\_covid, etc.):**  
different patterns for every season

Smoothed COVID Breakpoint Frequency across U.S. HSAs



**interpol\_frequency\_covid:** east and particularly southwest (near Tennessee, NC) have highest outbreak frequencies



**spatial\_coherence\_superimposed\_covid\_summer\_nohawaii,**  
**spatial\_coherence\_superimposed\_covid\_winter\_nohawaii:** more spatial coherence between east and west during summer outbreaks, less coherence and stronger diffusion overall during winter months